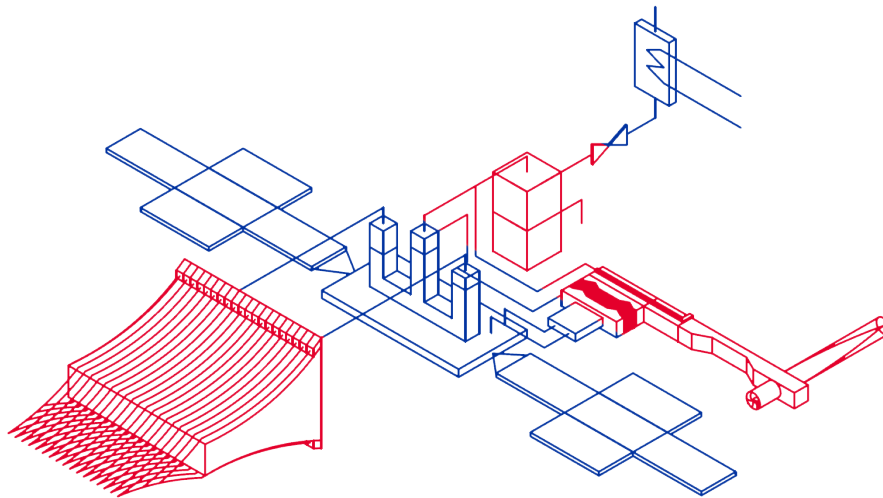


XSLR Aerospike Expander for Skyraker™ Technical Demo

Phase 3 — Cold Flow Test, Steam Thruster, and
Architecture Heritage Statement

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Phase 3 Case Study Overview

Phase 3A: Compressed N_2 propellant proxy — Validates nozzle geometry and thrust coefficient C_F , C_d

Phase 3B: Steam (distilled H_2O) — water-fed propulsion, I_{sp} measurement and comparison with CEA

Architecture target: Config. D+ hybrid EEC aerospike (full design documented in [?])

Nozzle: Aerospike, GFRP composite shell + PET GR foam core + Al throat insert (materials on hand)

Thrust measurement: 100 g load cell + lever arm + Arduino/HX711 **Steam generator:** 300 W cartridge heater, PID

Timeline: Construction start late June 2026 — Phase 3A target August–September 2026 — Phase 3B target October 2026

Phases 3A+3B combined budget: $\approx$ €300–€500

June 2026 (updated from March 2026)

Version 2.0

Part of the Space Framework™

Fact-checked: Sutton & Biblarz *RPE* 9th ed., Anderson *MCF* 3rd ed., NIST WebBook, NASA SP-8120.



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1 Overview, Scope, and Architecture Heritage

1.1 Position in the Skyraker Development Roadmap

The XSLR (eXperimental aeroSpike Liquid Rocket) is the chemical propulsion subsystem of Skyraker™ [1]. It converts LH₂/LOX produced by the SEA electrolysis assembly into high-thrust impulsive manoeuvres at $I_{sp,vac} = 465$ s and a throttle range of 12–100 % RPL.

Phase 3 of the development roadmap is the first experimental programme directly dedicated to XSLR. It is preceded by Phase 1 (WTP flash desalination, complete) and Phase 2 (SEA water electrolysis, complete) and is followed by Phase 4 (integrated chemical propulsion demonstration) and Phase 5 (ICER microwave plasma thruster and full-system integration).

Construction timeline update (June 2026): Phase 3 hardware construction starts late June 2026. Phase 3A (N₂ cold flow) is targeted for completion by September 2026. Phase 3B (steam thruster) is targeted for completion by October 2026. These dates supersede the original March 2026 Roadmap [1].

1.2 Architecture Heritage Statement

Phase 3 is explicitly a *subscale physical proxy* for the full-scale XSLR architecture. Table 1 maps each Phase 3 test element to its full-scale counterpart.

Table 1: Phase 3 subscale elements and their full-scale Config. D+ / Option C counterparts [?].

Phase 3 element	What it validates	Full-scale counterpart
N ₂ cold flow through composite aerospike nozzle	Nozzle geometry (C_F , C_d); thrust measurement system; composite structural integrity under gas loads	Aerospike plug contour; discharge coefficient at $p_c = 50$ bar
Steam propellant (distilled H ₂ O)	Water-fed propulsion principle; I_{sp} measurement; steam generator / PID control; condenser-free operation	Nozzle-loop LH ₂ thermal path in Config. D+ (the “nozzle loop” absorbs 154 kW at 22 kN and delivers heated gas to Turbine N)
Load cell + lever arm (100 g) + Arduino DAQ	Thrust measurement and data acquisition methodology	Future hot-fire load cell system at full scale
Composite nozzle (PET GR foam + E-glass + Al insert)	Fabrication process; thermal survivability in steam; surface quality for C_d	Not directly ported (full-scale uses regeneratively cooled Inconel/Ti); validates fabrication workflow

Known subscale gaps — declared, not corrected at this stage.

Three physical quantities of the full-scale Config. D+ / Option C architecture are *not* reproducible at Phase 3 scale and are listed here for transparency:

1. **Spike-to-nozzle heat-load ratio.** At full scale this rises to $\dot{Q}_s/\dot{Q}_n = 12.76$ due to the absolute size of the spike body ($L_{\text{plug}} \times W_{\text{span}}$ growing as k). At Phase 3 pressures (1–8 bar) and $r_t = 2$ mm the ratio is not measurable in proportion, since the spike is a single flat surface, not a live cooling channel.
2. **Bypass surplus fraction (71.2 %).** The 246 kW bypass at 22 kN — and 2065 kW at 220 kN — arise from the turbine power exceeding the pump demand. At Phase 3 there are no turbopumps; this fraction is not physically present. It is quantified analytically in [?].
3. **Dual-loop thermal coupling.** Phase 3 validates a single-path nozzle-loop proxy (steam). The spike loop and its electric pump / turbogenerator (Option C) have no Phase 3 counterpart; these are full-scale development items tied to DQ-A and DQ-D of [?].

2 Compressible Flow Theory

2.1 Isentropic Relations

For an ideal gas expanding from stagnation state (p_0, T_0) :

$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} M^2 \quad (1)$$

$$\frac{p_0}{p} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\gamma/(\gamma-1)} \quad (2)$$

$$\frac{A}{A_t} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2\right) \right]^{(\gamma+1)/[2(\gamma-1)]} \quad (3)$$

2.2 Choked Throat

At $M = 1$ the critical pressure ratio:

$$\frac{p^*}{p_0} = \left(\frac{2}{\gamma + 1}\right)^{\gamma/(\gamma-1)} = 0.5283 \quad (\gamma = 1.4, \text{ N}_2) \quad (4)$$

The throat chokes when $p_0/p_a > 1.893$, i.e. $p_c \geq 2p_a$. All Phase 3A set points satisfy this condition.

2.3 Thrust, Exit Velocity, and I_{sp}

$$F = \dot{m} v_e + (p_e - p_a) A_e \quad (5)$$

$$v_e = \sqrt{\frac{2\gamma}{\gamma - 1} \frac{RT_0}{M_{\text{mol}}} \left[1 - \left(\frac{p_e}{p_0}\right)^{(\gamma-1)/\gamma}\right]} \quad (6)$$

$$I_{\text{sp}} = \frac{F}{\dot{m} g_0} \quad (7)$$

2.4 Thrust Coefficient and Discharge Coefficient

$$C_F = \frac{F}{p_c A_t} \quad (8)$$

$$C_d = \frac{\dot{m}_{\text{meas}}}{\dot{m}_{\text{ideal}}}, \quad \dot{m}_{\text{ideal}} = p_0 A_t \sqrt{\frac{\gamma}{RT_0}} \left(\frac{2}{\gamma + 1} \right)^{(\gamma+1)/[2(\gamma-1)]} \quad (9)$$

2.5 Aerospike Altitude Compensation and I_{sp} Scaling

The aerospike auto-matches the effective expansion ratio ε_{eff} to ambient pressure, eliminating over-expansion losses across a wide altitude range. This makes it mission-enabling at Titan’s surface ($p_a = 146.7 \text{ kPa}$), where a bell nozzle at $\varepsilon = 58$ delivers $I_{\text{sp}} \approx 57 \text{ s}$ while the aerospike achieves $\approx 405 \text{ s}$ [?].

The gap between the Phase 3B steam I_{sp} and the full-scale LH_2/LOX value is given by the thermal scaling ratio:

$$\frac{I_{\text{sp,LH}_2/\text{LOX}}}{I_{\text{sp,steam}}} \approx \sqrt{\frac{T_{\text{c,LH}_2/\text{LOX}}/M_{\text{eff}}}{T_0^{\text{steam}}/M_{\text{H}_2\text{O}}}} = \sqrt{\frac{3520/10}{423/18}} \approx 3.87 \quad (10)$$

If Phase 3B measures $I_{\text{sp,steam}} \approx 70\text{--}78 \text{ s}$, the extrapolated prediction $I_{\text{sp,LH}_2/\text{LOX}} \approx 271\text{--}302 \text{ s}$ (thermal scaling); NASA CEA gives $\approx 430\text{--}465 \text{ s}$ at 50 bar (real-gas + combustion). The gap confirms that Phase 3 validates the propulsion *principle* and geometry, not the full thermodynamic performance.

Table 2: Predicted nozzle performance at Phase 3A test points ($\gamma = 1.4$, $T_0 = 293 \text{ K}$, N_2).

p_c [bar]	p_c/p_a	M_e	v_e [m/s]	N_2	$C_{F,\text{ideal}}$
2	2.0	1.0	315		1.000
3	3.0	1.6	368		1.290
4	4.0	1.9	400		1.420
5	5.0	2.2	425		1.510
7	7.0	2.5	455		1.580
10	10.0	2.9	480		1.630

3 Composite Nozzle Structure

3.1 Available Materials

Table 3: Available composite materials for XSLR nozzle fabrication.

Material	Specification	Quantity available
PET GR structural foam	80 kg/m ³ , 1025 × 2490 × 5 mm sheets	4 sheets ($\approx 10 \text{ m}^2$)
E-glass biaxial fabric	300 $\frac{\text{g}}{\text{m}^2}$, $\pm 45^\circ$, width 1270 mm	10 m ² (4 rolls)
Infusion epoxy	$T_g \approx 80\text{--}120^\circ \text{C}$	As needed

3.2 Material Properties

PET GR foam (80 kg/m³). Closed-cell PET structural foam. Compressive strength $\sigma_c \approx 1.4 \text{ MPa}$; shear strength $\tau \approx 0.8 \text{ MPa}$; thermal stability to $\approx 100^\circ \text{C}$ continuous (short-term: 120°C). Excellent machinability; surface finish adequate for direct lamination.

E-glass biaxial $\pm 45^\circ$, 300 g/m². The $\pm 45^\circ$ orientation maximises in-plane shear stiffness and torsional rigidity — ideal for a tubular aerospike shell under internal pressure and bending. Typical laminate properties (single ply infused, FVF 50 %): $E_{11} \approx E_{22} \approx 16$ GPa, $G_{12} \approx 8$ GPa, $\sigma_{\text{tens}} \approx 250$ MPa.

3.3 Quick Structural Check

Internal pressure load at $p_c = 5$ bar, throat annulus $D_t = 6$ mm, wall thickness $t = 2$ mm (two E-glass plies):

$$\sigma_{\text{hoop}} = \frac{p_c r}{t} = \frac{5 \times 10^5 \times 0.003}{0.002} = 0.75 \text{ MPa} \ll \sigma_{\text{tens}} \approx 250 \text{ MPa} \quad \checkmark \quad (11)$$

Safety factor > 300 ; composite shell is structurally over-specified. Primary pressure boundary is the SS316 combustion chamber.

3.4 Spike Contour

Parabolic approximation for the laboratory demonstrator:

$$r(x) = r_t - \frac{r_t - r_{\text{tip}}}{L_s^2} x^2, \quad 0 \leq x \leq L_s \quad (12)$$

Table 4: XSLR demonstrator aerospike design parameters.

Parameter	Symbol	Value	Unit
Design pressure ratio	p_c/p_a	5	–
Design exit Mach	M_e	2.2	–
Throat radius	r_t	2	mm
Throat area	A_t	12.57	mm ²
Expansion ratio	ε	5	–
Spike length	L_s	40–60	mm
Tip radius	r_{tip}	1.5	mm
Initial half-angle	θ_i	25	°
Spike truncation	L/L_{ideal}	30	%

3.5 Fabrication Sequence

- F1. Foam machining.** CNC-route or hand-carve PET GR foam to spike contour profile (Eq. (12)). Sand to 120 grit; wipe with acetone.
- F2. Wet layup.** Cut two plies of $\pm 45^\circ$ E-glass to spike template. Apply infusion epoxy; roll out over foam mandrel, removing air bubbles.
- F3. Vacuum bag and cure.** Apply peel ply + perforated release + breather + bag. Pull vacuum to ≥ 0.9 bar relative. Cure 24 h at RT, post-cure 4 h at 60°C if oven available.
- F4. Aluminium throat insert.** Machine a short Al6061 ring (OD = throat annulus OD, length 20 mm). Bore annular throat to exact r_t dimension on the lathe.
- F5. Bond insert.** Epoxy-bond Al insert into composite nozzle at throat station using high-temperature adhesive ($T_{\text{max}} \geq 200^\circ\text{C}$ for Phase 3B compatibility). Cure under clamp pressure.
- F6. Finish.** Sand exterior to 400 grit; coat with thin epoxy gel coat to seal porosity and improve C_d .

4 Hardware — Phase 3A (Cold Flow)

Table 5: Phase 3A hardware list.

Component	Spec	Notes	Cost
N ₂ cylinder	5 L, 200 bar	Food/technical grade; refillable	≈€50 dep.
Pressure regulator	0–10 bar, $\frac{1}{4}$ BSP	Two-stage preferred	≈€30
Needle valve	SS316, $\frac{1}{4}$ BSP	Fine flow control	≈€12
Pressure transducer	0–10 bar, 0–5 V	Calibrate before test	≈€15
SS316 tubing	6 mm OD, 1 mm wall	Rated >100 bar burst	≈€20
Safety relief valve	12 bar, $\frac{1}{4}$ BSP	Mandatory	≈€15
Combustion chamber	SS316, custom, 3 mm wall	CNC at fab lab	≈€40
XSLR composite nozzle	PET GR + E-glass + Al insert	See § 3	≈€20
Load cell	100 g full-scale, ±0.1 g	Pre-load calibrate	≈€8
HX711 ADC	24-bit	Gain = 128 default	≈€3
Arduino Uno	ATmega328P	10 Hz sampling	≈€25
Lever arm	Al profile, 15–20 cm	Pivot on knife edge	≈€10
Rotameter	0–10 L/min N ₂	Mass flow proxy	≈€20
PTFE tape + fittings	–	3–4 wraps per joint	≈€10
Phase 3A total			≈€278

Lever arm sizing. At $p_c = 5$ bar, $r_t = 2$ mm: $F = C_F \cdot p_c \cdot A_t = 1.51 \times 5 \times 10^5 \times 12.57 \times 10^{-6} \approx 9.5$ N. A 100 g load cell (full scale ≈ 0.98 N) requires $\ell_2/\ell_1 \geq 9.5/0.98 \approx 10$. Use $\ell_1 = 3$ cm, $\ell_2 = 30$ cm; $F_{\text{cell}} = F/10 \approx 0.95$ N — within full scale.

5 Hardware — Phase 3B (Steam Thruster)

Table 6: Phase 3B hardware list (incremental over Phase 3A).

Component	Spec	Notes	Cost
Steam generator vessel	SS316, $\varnothing 50$ mm, L=120 mm, 3 mm wall	Fab lab CNC; burst >150 bar	≈€60
Cartridge heater	300 W, 220 V, SS316 sheath, 6 mm dia.	Never operate dry	≈€15
SSR-40DA relay	3–32 V DC in, 40 A AC out	Mount on heatsink	≈€8
PID controller	12 V, K-type thermocouple input	Set $T_{\text{target}} = T_{\text{sat}} + 10$ K	≈€12
K-type thermocouples ×2	0–400 °C, M6 thread	Cross-check before test	≈€10
Back-pressure regulator	0–10 bar, SS316	Maintains steady p_c	≈€30
Safety relief valve	12 bar, $\frac{1}{4}$ BSP	Test function every session	≈€15
Digital scale	0.1 g, ≥500 g capacity	Gravimetric $\dot{m}_w$	≈€12
Peristaltic pump (optional)	12 V, 0.1–10 mL/min	Precise flow control	≈€25
Thermal insulation	Ceramic wool, $T_{\text{max}} > 500$ °C	Wrap generator and lines	≈€10
PPE	Insulated gloves 200 °C + face shield	Mandatory for steam ops	≈€15
All Phase 3A hardware	–	Reused without modification	€0
Phase 3B incremental cost			≈€212
Phase 3A+3B combined			≈€490

6 Steam Thermodynamics

6.1 Steam Propellant Properties

For superheated steam at 120–200 °C, the ideal gas approximation with $\gamma = 1.30$, $R = 461.5$ J/(kg K) is accurate to <3 % (NIST):

$$v_e^{\text{steam}} = \sqrt{\frac{2 \times 1.30}{0.30} \times 461.5 \times T_0 \left[1 - \left(\frac{p_e}{p_0} \right)^{0.30/1.30} \right]} \quad (13)$$

At $T_0 = 423$ K, $p_c = 5$ bar, $p_a = 1$ bar: $v_e \approx 760$ m/s, $I_{\text{sp,steam}} \approx 77.5$ s.

6.2 Heater Power Budget

$$P_{\text{heater}} = \dot{m}_w [c_{p,L}(T_{\text{sat}} - T_{\text{in}}) + \Delta h_{\text{vap}}(T_{\text{sat}}) + c_{p,V}(T_0 - T_{\text{sat}})] \quad (14)$$

At $\dot{m}_w = 1$ g/s, $p_c = 5$ bar ($T_{\text{sat}} = 151.8$ °C), $T_0 = 160$ °C: $P \approx 267$ W. A 300 W heater is sufficient.

Table 7: Steam thruster thermodynamic parameters (NIST steam tables).

p_c [bar]	T_{sat} [°C]	Δh_{vap} [J/g]	P [W] ($\dot{m} = 1$ g/s)	v_e [m/s]	I_{sp} [s]
1.0	100.0	2257	255	545	55.6
2.0	120.2	2201	260	600	61.2
3.0	133.5	2163	263	660	67.3
5.0	151.8	2107	267	760	77.5
8.0	170.4	2047	273	840	85.6

6.3 Phase 3B to Full-Scale: Subscale vs. Full-Scale Comparison

Table 8: Key XSLR parameters across development stages. Invariant thermodynamic quantities in italics.

Parameter	Phase 3B demo	XSLR-40	XSLR-220	XSLR-250
F_{vac} (kN)	$\lesssim 0.01$	40–47	220	250
p_c (bar)	1–8	50	50	50
Propellant	H ₂ O steam	LH ₂ /LOX	LH ₂ /LOX	LH ₂ /LOX
r_t (mm)	2	9.9	23.7	24.7
L_{plug} (cm)	4–6	61.8	118.9	154.5
W_{span} (cm)	5	98.6	118.6	246.6
C_F (<i>ideal</i> , $p_c/p_a = 5$)	<i>1.51</i>	<i>1.51</i>	<i>1.51</i>	<i>1.51</i>
$I_{\text{sp,vac}}$ (s)	55–86	465	465	465
Architecture	Subscale proxy	D+/Opt.C	D+/Opt.C	D/Opt.C

7 Experimental Procedures

7.1 Phase 3A — Pre-Test and Cold Flow Procedure

- 3A-P1. Nozzle inspection.** Check composite skin for delamination; verify Al insert bond integrity. Remove burrs at throat with 600-grit sandpaper.
- 3A-P2. Leak test.** Pressurise to $p_{c,\text{max}}$ with N₂; apply soap solution to all fittings; pressure drop < 0.05 bar in 2 min = pass.
- 3A-P3. Load cell calibration.** Tare with nozzle mounted; apply known masses 10, 20, 50 g; fit linear calibration $F [\text{mN}] = k \times \text{ADC}$.
- 3A-P4. Open N₂** to first set point; allow 10 s stabilisation.
- 3A-P5. Record 30 s** at 10 Hz: p_c , F_{cell} , $\dot{V}_{\text{N}_2}$, T_0 .
- 3A-P6. Compute:** $F = F_{\text{cell}} \cdot (\ell_2/\ell_1)$; $C_F = F/(p_c A_t)$; $C_d = \dot{m}_{\text{meas}}/\dot{m}_{\text{ideal}}$.
- 3A-P7. Increment pressure** to next set point (Table 2). Run 3 replicates per point.
- 3A-P8. Shut down:** close cylinder valve, vent line, disconnect.

7.2 Phase 3B — Steam Warm-Up and Firing Procedure

- 3B-P1. Fill steam generator** to 60–70 % with WTP distillate (Phase 1 output, $C_d < 200 \mu\text{S}/\text{cm}$).
- 3B-P2. Cold leak test** with 5 bar N₂ on steam circuit; drain N₂.

3B-P3. Relief valve test: verify opens at 12 bar; reset.

3B-P4. Thermocouple check: both sensors in boiling water; accept $100 \pm 2^\circ\text{C}$.

3B-P5. Weigh reservoir. Record m_0 on digital scale.

3B-P6. Power heater. PID set to $T_{\text{sat}}(p_c) + 10\text{ K}$. Close back-pressure regulator. Wait for stable $T_0 \pm 2^\circ\text{C}$, $p_c \pm 0.1\text{ bar}$ (5–10 min).

3B-P7. Open back-pressure regulator to target p_c ; fine-tune needle valve.

3B-P8. Start Arduino DAQ; record 60 s at 10 Hz.

3B-P9. Close nozzle valve. Weigh reservoir: record m_1 . $\dot{m}_w = (m_0 - m_1)/\Delta t$.

3B-P10. Compute: $I_{\text{sp,meas}} = F_{\text{nozzle}}/(\dot{m}_w g_0)$; $\eta_{I_{\text{sp}}} = I_{\text{sp,meas}}/I_{\text{sp,theor}}$.

3B-P11. Cool down to $<50^\circ\text{C}$ before disassembly.

8 Data Acquisition Tables

Table 9: Phase 3A data acquisition sheet. ($r_t = 2\text{ mm}$, $A_t = 12.57\text{ mm}^2$, $\ell_2/\ell_1 = 10$, $T_0 \approx 20^\circ\text{C}$.)

p_c [bar]	$\dot{m}_{\text{ideal}}$ [g/s]	F_{ideal} [N]	F_{meas} [N]	$\dot{m}_{\text{meas}}$ [g/s]	C_F	C_d
2	0.52	0.82				
3	0.78	1.47				
4	1.04	2.09				
5	1.30	2.60				
7	1.82	3.78				
10	2.60	5.62				

Table 10: Phase 3B data acquisition sheet. ($r_t = 2\text{ mm}$, $A_t = 12.57\text{ mm}^2$, $\varepsilon = 5$, $\ell_2/\ell_1 = 10$.)

p_c	T_{sat}	$I_{\text{sp,th}}$	F_{ideal}	F_{meas}	$\dot{m}_w$	$I_{\text{sp,meas}}$	$\eta_{I_{\text{sp}}}$
[bar]	[$^\circ\text{C}$]	[s]	[N]	[N]	[g/s]	[s]	
1.0	100.0	55.6	0.35				
2.0	120.2	61.2	0.77				
3.0	133.5	67.3	1.28				
5.0	151.8	77.5	2.45				
8.0	170.4	85.6	4.30				

9 Safety

9.1 Phase 3A — Compressed N₂

N₂ cylinder service pressure: up to 200 bar.
 N₂ is an asphyxiant: >75 % v/v causes unconsciousness without warning.
 Fitting failure at pressure is a projectile hazard.

Applicable standards: EN ISO 14246; EIGA Doc. 33; PED 2014/68/EU.

- (a) Ventilation ≥ 6 air changes/hour; O₂ deficiency monitor recommended (alarm <19.5 % O₂).
- (b) Safety relief valve at 12 bar: mandatory.
- (c) All fittings rated ≥ 30 bar ($3\times$ working pressure).
- (d) Polycarbonate blast shield (≥ 6 mm) during first firing.
- (e) Two-person rule; cylinder chained to wall.
- (f) Depressurise before disconnecting: close valve, vent line, disconnect.

9.2 Phase 3B — Steam

Steam at 150 °C carries ≈ 2100 J/g latent heat — burns are more severe than boiling water at the same temperature.
 220 V heater in a wet environment: shock hazard.
 Fitting failure under pressure + steam: projectile + scald hazard.

Applicable standards: PED 2014/68/EU; EN 13445; ISO 4126-1; CEI 64-8.

- (a) Safety relief valve at 12 bar: mandatory; test every session.
- (b) 220 V heater connected via RCD ($I_{\Delta} \leq 30$ mA); SSR in fully enclosed box.
- (c) Never operate heater with empty vessel.
- (d) PPE: insulated gloves ≥ 200 °C, face shield, lab coat.
- (e) Two-person rule.
- (f) Cool to <50 °C and $p = 0$ before disassembly.

10 Phase 3 Completion Criteria

Phase 3A pass: C_F within 15 % of ideal for at least 4 pressure set points; $C_d \geq 0.85$; repeatability <10 % over 3 replicates; no structural damage to composite nozzle after full test campaign. Results documented as subscale anchor for the C_F/C_d model used in Config. D+/Option C full-scale design.

Phase 3B pass: $F > 0$ mN measured with load cell at all 5 steam pressure points; $I_{sp,meas}$ within 25 % of CEA steam prediction; scaling ratio $I_{sp,meas}/I_{sp,theor} \geq 0.75$; repeatability <10 % over 5 successive firings; nozzle survives full thermal cycle without delamination. $I_{sp,meas}$ recorded as input to Eq. (10) for extrapolation toward LH₂/LOX full-scale.

11 Key Equations Reference

Table 11: Complete equation reference for XSLR Phase 3.

Quantity	Equation	Eq.
Isentropic temperature	$T_0/T = 1 + (\gamma - 1)M^2/2$	(1)
Isentropic pressure	$p_0/p = [1 + (\gamma - 1)M^2/2]^{\gamma/(\gamma-1)}$	(2)
Area–Mach	$A/A_t = (1/M)[\dots]^{\gamma/2(\gamma-1)}$	(3)
Critical pressure	$p^*/p_0 = 0.5283 \text{ (N}_2\text{)}$	(4)
Thrust	$F = \dot{m}v_e + (p_e - p_a)A_e$	(5)
Exit velocity	$v_e = \sqrt{2\gamma RT_0/[(\gamma - 1)M_{\text{mol}}](1 - \dots)}$	(6)
I_{sp}	$I_{\text{sp}} = F/(\dot{m}g_0)$	(7)
Thrust coefficient	$C_F = F/(p_c A_t)$	(8)
Discharge coefficient	$C_d = \dot{m}_{\text{meas}}/\dot{m}_{\text{ideal}}$	(9)
I_{sp} scaling to LH ₂ /LOX	$I_{\text{sp,LH}_2/\text{LOX}}/I_{\text{sp,steam}} \approx 3.87$	(10)
Spike contour	$r(x) = r_t - (r_t - r_{\text{tip}})x^2/L_s^2$	(12)
Heater power	$P = \dot{m}_w[c_{p,L}\Delta T_{\text{sub}} + \Delta h_{\text{vap}} + c_{p,V}\Delta T_{\text{sup}}]$	(14)
Steam exit velocity	$v_e^{\text{steam}} = \sqrt{2\gamma RT_0/(\gamma - 1)[1 - \dots]}$	(13)

References

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